I don't think you will get these in your 11+ paper, but you may in other entrance exams. It is good fun to know how to do them because they are easy when you know how.

Multiplication of fractions

Rule 1: **of** means **times.** So: $\frac{2}{5}$ **of** $\frac{1}{2}$ means $\frac{2}{5} \times \frac{1}{2}$

Rule 2: **Cross cancel** if you can. Examples:

Here I have cancelled out the 2s. This is called **cross cancelling.**

$$\frac{\cancel{2}^{\,1}}{5} \times \frac{1}{\cancel{2}_{\,1}} = \frac{1}{5}$$

In this second example I have reduced the 9 to 3 and the 6 to 2 by dividing each by 3. This is also called **cross cancelling.**

$$\frac{\cancel{9}^{\,3}}{11} \times \frac{4}{\cancel{6}_{\,2}} = \frac{\cancel{12}^{\,6}}{\cancel{22}_{\,11}} = \frac{6}{11}$$

Always answer in lowest terms

When you have cross cancelled you simply multiply the two top numbers (numerators) and then the two bottom numbers (denominators).

Can you see how this works? Now try these:

1. $\frac{6}{7}$ of $\frac{2}{3}$ =
2. $\frac{8}{9} \times \frac{3}{5}$ =
3. $\frac{10}{1} \times \frac{2}{15}$ =
4. $\frac{11}{12} \times \frac{4}{7}$ =
5. $\frac{1}{5} \times \frac{5}{6}$ =
6. $\frac{3}{4}$ of $\frac{8}{9}$ =

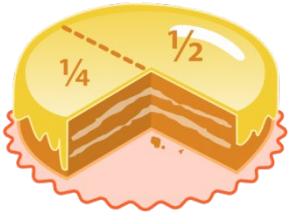
You will need to change the rest into improper fractions first. Then carry on as normal:

7. $3\frac{3}{1} \times 5\frac{5}{6}$ =
8. $6\frac{3}{4} \times 2\frac{2}{9}$ =
9. $3\frac{3}{8} \times 5\frac{1}{3}$ =
10. $3\frac{1}{6} \times 1\frac{4}{5}$ =

Did you enjoy those?

Fractions – Order of Size

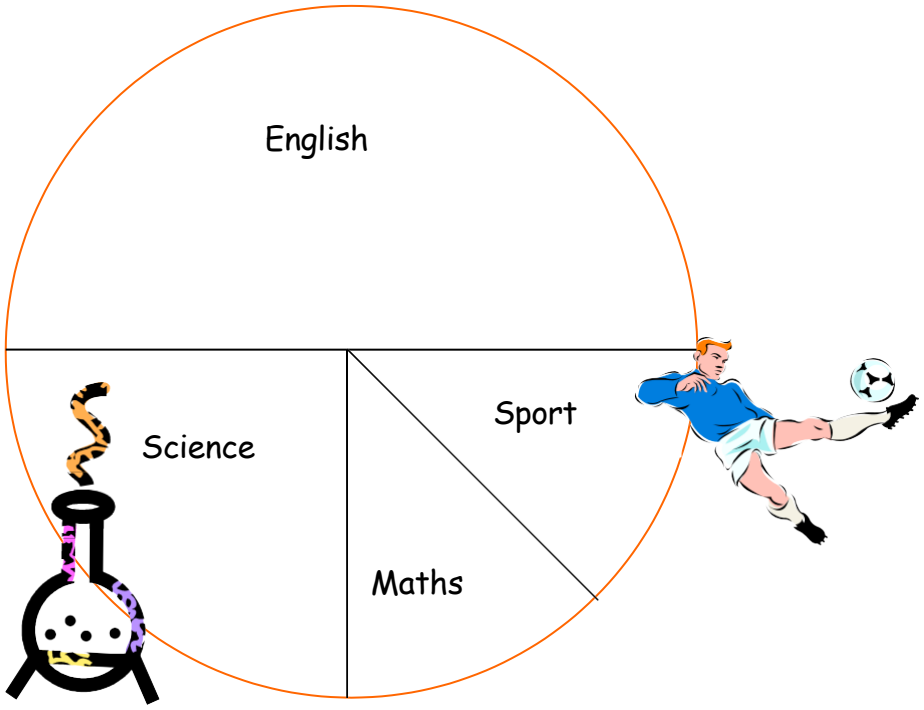
- You have 4 fractions to order from the smallest to the largest. This can be difficult to do with different denominators.
- Use the first column to change all the fractions to a common denominator.
- Then write your answer in the second column using the original fractions.



		Fractions with common denominator	Order of fractions (Smallest to Largest)
1.	$\frac{3}{8}$ $\frac{1}{2}$ $\frac{1}{3}$ $\frac{1}{6}$	$\frac{9}{24}$ $\frac{12}{24}$ $\frac{8}{24}$ $\frac{4}{24}$	$\frac{1}{6}$ $\frac{1}{3}$ $\frac{3}{8}$ $\frac{1}{2}$
2.	$\frac{3}{5}$ $\frac{3}{4}$ $\frac{2}{3}$ $\frac{7}{12}$		
3.	$\frac{1}{2}$ $\frac{3}{4}$ $\frac{7}{12}$ $\frac{4}{6}$		
4.	$\frac{2}{5}$ $\frac{3}{8}$ $\frac{1}{4}$ $\frac{3}{10}$		
5.	$\frac{2}{3}$ $\frac{1}{6}$ $\frac{2}{9}$ $\frac{1}{3}$		
6.	$\frac{1}{4}$ $\frac{1}{8}$ $\frac{1}{3}$ $\frac{9}{24}$		
7.	$\frac{2}{5}$ $\frac{1}{3}$ $\frac{7}{15}$ $\frac{4}{15}$		
8.	$\frac{3}{4}$ $\frac{5}{6}$ $\frac{9}{10}$ $\frac{2}{3}$		

Did you find this one easy? Remember by changing the denominators of fractions you can make your life a whole lot easier.

Percentages - Favourite Subjects



1. What <u>fraction</u> of the children like each subject?

English: _____ Sport: _____

Maths: _____ Science: _____

2. What <u>percentage</u> like each subject?

English: _____ Sport: _____

Maths: _____ Science: _____

3. If there are 40 children in the class, how many like:

a) English? _____

b) Sport? _____

c) Maths? _____

d) Science? _____

Simple Equations

Remember

4x = 16 (4x means 4 times x)
The value of x = 4. You could divide 16 by 4 to find the value of x.

x + 5 = 18. To find x you take away 5 from 18 = 13.

x - 1 = 19. To find x you add 1 to 19 = 20.

x ÷ 2 = 18. To find x you times 18 by 2 = 36.

Now try these – Find x.

	Answers		Answers
3 + x = 12	-	x + 5 = 9	-
x + 8 = 9	-	$\frac{x}{5}$ = 3	-
4 x = 20	-	x + 8 = 16	-
15 – x = 7	-	10 – x = 5	-
x – 4 = 1	-	x + 8 = 22	-
x ÷ 2 = 5	-	x + 7 = 13	-
5 x = 25	-	11 – x = 5	-
$\frac{x}{3}$ = 5	-	x + 3 = 4	-
3 x = 9	-	x + 3 = 5	-
X – 1 = 5	-	x + 6 = 11	-
8 x = 16	-	14 – x = 7	-

SCORE

	22

Equations Game

Cut out the cards and match each equation with the correct value of x

$x=6$	$3+x=12$	$x=5$	$5x=25$
$x=4$	$x+8=9$	$x=5$	$\frac{x}{3}=5$
$x=15$	$4x=20$	$x=9$	$3x=9$
$x=5$	$15-x=7$	$x=5$	$x-1=5$
$x=10$	$x-4=1$	$x=12$	$8x=16$
$x=3$	$x\div2=5$	$x=1$	$x+5=9$

Equations Game

Cut out the cards and match each equation with the correct value of x

$x=8$	$\frac{x}{5}=3$	$x=2$	$x\div3=4$
$x=15$	$x+8=16$	$x=8$	$x+3=5$
$x=16$	$10-x=5$	$x=6$	$x+6=11$
$x=4$	$x\div8=2$	$x=2$	$x-3=1$
$x=6$	$4x=16$	$x=5$	$14-x=7$
$x=4$	$11-x=5$	$x=7$	$x+7=13$

11 PLUS PREPARATION

MATHS PAPER

Name:	
Date of Birth:	
Today's Date:	

READ THESE INSTRUCTIONS CAREFULLY BEFORE STARTING.

- You have 35 minutes to complete the paper.
- Working out can be done on the paper or on spare paper but ensure only your answer is written clearly on the answer line.
- Each question is worth one mark. If you can't do a question, then move on.
- If you finish early then check your answers.
- Once the test has started you may not ask for help.
- No calculators should be used.

1. Calculate: 347 x 6 _____

2. Calculate: 3 2/3 – 2 5/6 _____

3. Calculate: 4/5 of 120 _____

4. Calculate: 2.8 + 5.85 _____

5. Calculate:

a. 135.6 ÷ 100 _____

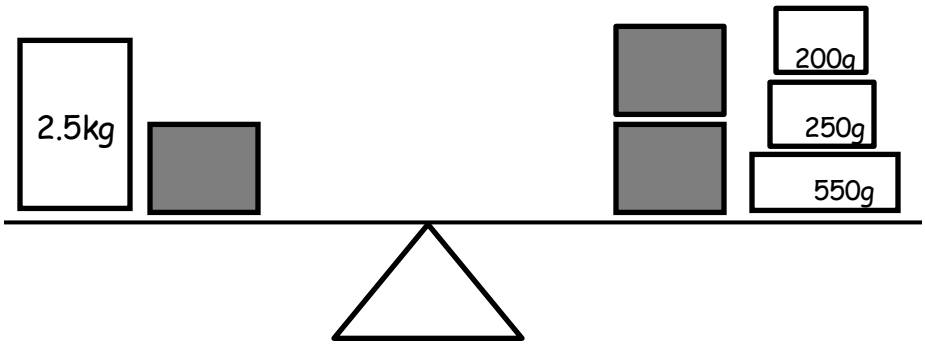
b. 2.7 ÷ 10 _____

c. 6 1/3 + 3 2/6 _____

d. 429 x 25 _____

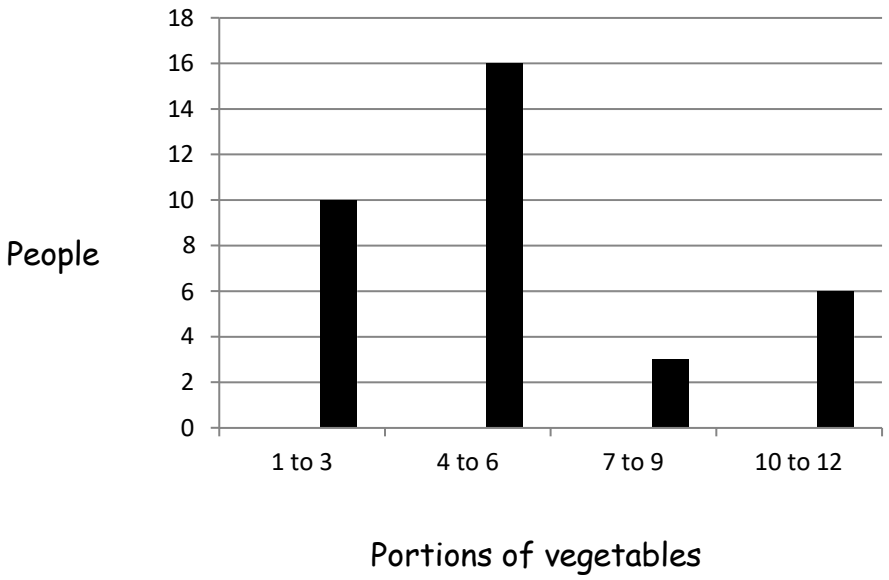
6. 5cm on a map represents 1km on the ground. What is the distance on the ground between two churches which are 19 cm apart on the map? _____

7.



The three shaded boxes each weigh the same.
How much does one shaded box weigh? _____

8.



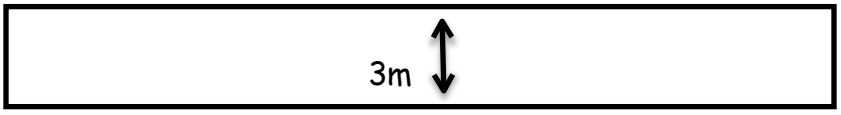
This graph shows the number of portions of vegetables eaten by people in a week.

a/ How many people ate between 7 and 9 portions per week? _____

b/ How many people were surveyed? _____

9. Janet used her calculator for a sum and the answer was shown as 37.489. Write down this number rounded to the nearest whole number. _____

10. A water feature in the grounds of a large house is 3 metres wide and rectangular in shape. How long is the water feature if it covers an area of 36 square metres? _____



11. A glass holds a quarter of a litre. How many glasses can be completely filled from a bottle holding 13.95 litres of water? _____

12. 1 litre of water is used to fill 4 smaller bottles of the same size. How many millilitres will there be in each bottle? _____

13. Philip's large watering can weighs 21kg when full of water. After he pours out half of the water it weighs 12kg.

What is the weight of the empty watering can? _____

14. Which number is closest to zero? _____

A. 0.202 B. 0.02 C. 0.022 D. 0.0022 E. 0.22

15. The school football team played 20 games. It won 40% of these and drew 20%.

a/ What percentage did it lose? _____

b/ How many games did it win? _____

c/ What would their win/loss ratio have been? _____

16. Which of these is not the same as 20%? _____

a/ 1/5

b/ 7/10 + 2/20

c/ 2.0 percent

d. 20/100

e. 0.20

17. A school hockey match starts at 3.45pm. It has two halves of 35 minutes each and a half time break of 8 minutes. At what time does it end? _____

18. A block of stone measuring 2 metres long, 1 metre wide and 80 cm high is cut into cubes of side 10cm.

How many cubes of stone will there be? _____

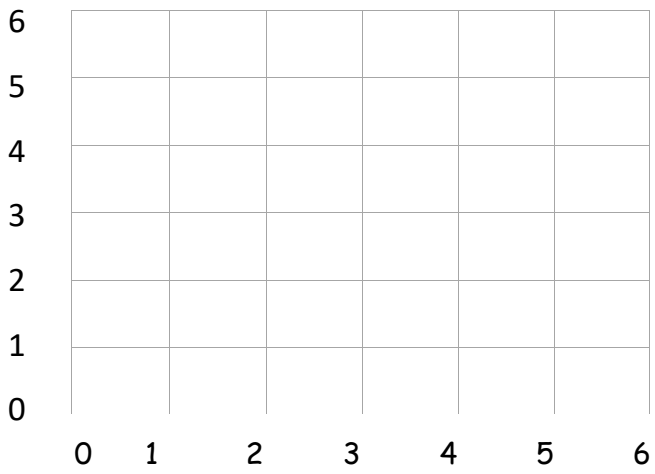
19. In the vegetable garden, Peter's father is planting a row of lettuces. He has planted the first and last lettuces in the row and they are 2 metres apart.
If lettuces have to be planted 5cm apart, how many lettuces should he plant between the first and the last lettuces in order to complete the row?

_____

In another part of the garden Peter has three apple trees of different maturity. One produces 22 apples, one produces 39 apples and the third produces 47 apples.

What is the average number of apples one tree produces? _____

20. a/ On the axes below plot the points:
A (0,1), B (2,4), C (4,1), D (6,4)
and join them up to make a 4-sided shape ABCD.



b/ How many lines of symmetry does the shape that you have drawn have? _____

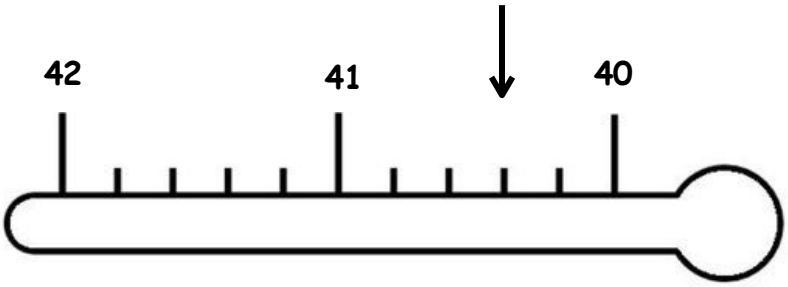
21. On a cold night in the Alps the temperature fell from -3°C to -15°C. How many degrees did it fall? _____

22.

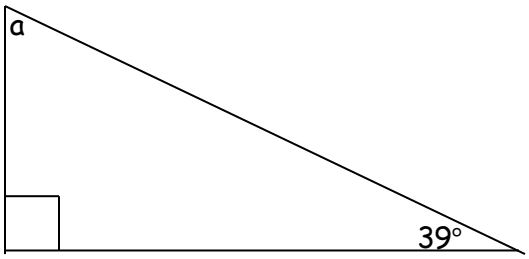
On a certain night the temperature in Greenland fell by 42°C from the daytime maximum of 7°C.

What was the night time temperature? _____

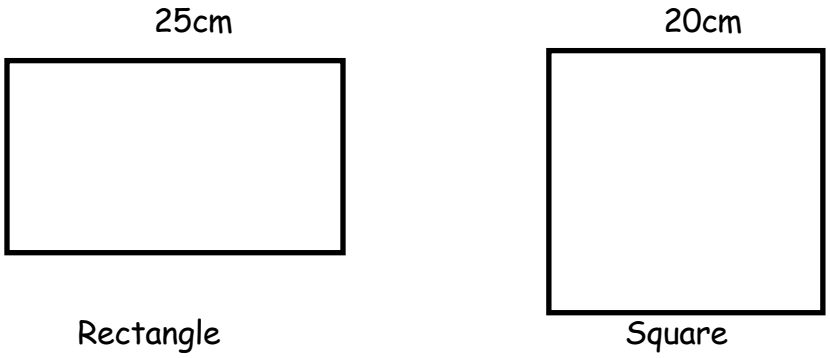
23. What temperature is shown on this thermometer? _____



24. Calculate the value of a. _____



25. The rectangle and the square both have a perimeter of 80 cm.



What is the area of the larger shape? _____

26. On Hugo's deer farm there are 394 deer. There are 246 adult deer and 320 hinds and calves.

(Stag = male adult, hind = female adult).

How many stags are there on the farm? _____

---

27. It takes 3 carpenters 3 days to make 3 wooden window frames.

a. How long would it take 3 carpenters to make 1 window frame? _____
b. How long would it take 1 carpenter to make 1 window frame? _____

The three carpenters were joined by two others and turned their hand to boat building. It took them all 12 months to build three identical boats.

c. How long would it have taken them all to build one boat? _____
d. How long would it have taken the original three carpenters to build five boats? (give your answer to the nearest month rounded down) _____

---

28. Harry takes six minutes to run 2000 metres.

a. If he ran at the same speed, how long would it take him to run 1km? _____

b. What speed did he run at in km/h _____ ?

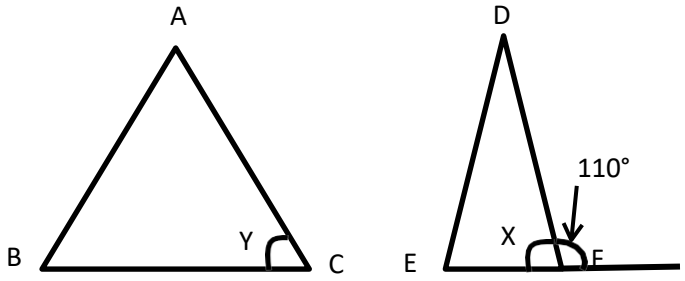
c. If 1km/h = 0.6mph, what was his speed in mph? _____

---

29. Paula thinks of two consecutive numbers. If she adds them she gets an answer between 10 and 14, if she multiples them she gets and answer between 40 and 45.

What are the two numbers? _____

30. Look at the 2 triangles below, the ABC is equilateral; the triangle DEF is isosceles with sides with sides DE and DF being the same length.



a. By how many degrees is angle X greater than angle Y? _____

31. If the temperature in Kabul is -7°c and the temperature in Vienna is 12°c what is the difference between the two temperatures? _____

- -

32. Jim's watering can weighs 25kg when full of water. After he pours out half of the water it weighs 13kg.

What is the weight of the empty watering can? _____

- -

33. At St. Joseph's sports academy there are 100 pupils. Sixty play rugby and twenty-five play cricket.

If fourteen pupils play both rugby and cricket, how many play neither? _____-

- -

34. A latte is made from milk and water. There is four times as much milk as water in the drink.

How much water would be needed to make 30 litres of latte? _____

_____END OF TEST _____

If you have time

- Go back and complete any questions you missed out.

- Check your answers carefully.

CORE SKILLS 8

1

This table shows the results for a polo team. How many matches did the team not win?

Year	Won	Lost	Drawn
2014	11	4	5
2015	6	5	9
2016	7	7	6

2

A number is made up of 8 units, 9 tens and 4 hundreds. Mary divides this number by 10.
What answer should Mary get?

489 894 498 49.8 48.9

3

Which is the lowest number that can be made by using 7 and 12 as factors?

4

54 children are going on a trip to the seaside. Half of the children take y pounds for spending on the arcade machines, the other half take z pounds for surfing lessons. All of them take x pounds for ice cream. What is the total amount, in pounds, taken by the group?

5

Peter has to fill flower vases with water from a can. The can holds $3\ ^{1}/_{3}$ litres of water. Each vase must have 330ml of water added to it.

How many vases of water can Peter fill from one can?

10 vases 12 vases 13 vases 11 vases 9 vases

6

Look at this statement.
85 is ? % of 238. A number is missing.

What number completes the statement?

34 36 $35\ ^{7}/_{10}$ $30\ ^{1}/_{2}$ $33\ \frac{1}{2}$

CORE SKILLS 8

7

A music festival lasting for a week recorded the number of number of people who attended (in hundreds) over a two year period.

	Year 1							Year 2						
Day	Mon	Tue	Wed	Thu	Fri	Sat	Sun	Mon	Tue	Wed	Thu	Fri	Sat	Sun
No.	21	19	24	19	23	30	28	19	15	21	15	23	28	26

What is the mode?

8

Look at this calculation.

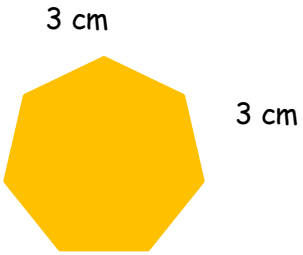
7 x 487 = 3409

What is the answer to the calculation 70 x 487?

34009 30409 34090 03409 43009

9

This is the plan of a desk. The scale of the drawing is 1:28.

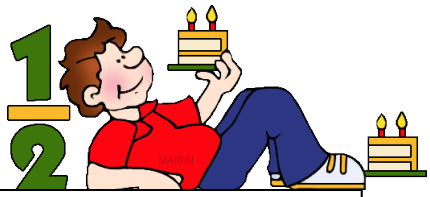


What is the full measurement around the desk?

10

In 2012 Sarah bought a washing machine, it was on sale at 35% off the original price of £260. She sold it in 2016 for a quarter of what she paid for it. How much change did the new owner receive out of £50?

Maths Reasoning



These questions are designed to get your mind whirring - so have a go. If you can't do it first time then try, try again. It is important to persevere.

At what time of what day of what year will it be _____ after the start of the year 2020?

a) 2020 seconds

b) 2020 minutes

c) 2020 hours

d) 2020 days

e) 2020 weeks

You may use a calculator for this one if you get stuck, but have a go at it without first!

MR 24

ANSWERS - COMPLETE MATHS – PART 39

Maths

Multiplication of Fractions

1. $\frac{\cancel{6}^{2}}{7}$ of $\frac{2}{\cancel{3}_{1}} = \frac{4}{7}$

2. $\frac{8}{\cancel{9}_{3}} \times \frac{\cancel{3}^{1}}{5} = \frac{8}{15}$

3 $\frac{\cancel{10}^{2}}{11} \times \frac{2}{\cancel{15}_{3}} = \frac{4}{33}$

4. $\frac{11}{\cancel{12}_{3}} \times \frac{\cancel{4}^{1}}{7} = \frac{11}{21}$

5 $\frac{1}{\cancel{5}_{1}} \times \frac{\cancel{5}^{1}}{6} = \frac{1}{6}$

6. $\frac{\cancel{3}^{1}}{\cancel{4}_{1}}$ of $\frac{\cancel{8}^{2}}{\cancel{9}_{3}} = \frac{2}{3}$

Did you cross cancel twice in question 6?

7. $3\frac{3}{1} \times 5\frac{5}{6} = \frac{\cancel{33}^{11}}{\cancel{10}_{2}} \times \frac{\cancel{35}^{7}}{\cancel{6}_{2}} = \frac{77}{4} = 19\frac{1}{4}$

8. $6\frac{3}{4} \times 2\frac{2}{9} = \frac{\cancel{27}^{3}}{\cancel{4}_{1}} \times \frac{\cancel{20}^{5}}{\cancel{9}_{1}} = \frac{15}{1} = 15$

9. $3\frac{3}{8} \times 5\frac{1}{3} = \frac{\cancel{27}^{9}}{\cancel{8}_{1}} \times \frac{\cancel{16}^{2}}{\cancel{3}_{1}} = \frac{18}{1} = 18$

10. $3\frac{1}{6} \times 1\frac{4}{5} = \frac{19}{\cancel{6}_{2}} \times \frac{\cancel{9}^{3}}{5} = \frac{57}{10} = \frac{7}{10}$

Order of fraction sizes

1. Correct order of size is: $\frac{1}{6}$ $\frac{1}{3}$ $\frac{3}{8}$ $\frac{1}{2}$

2. Correct order of size is: $\frac{7}{1}$ $\frac{3}{5}$ $\frac{2}{3}$ $\frac{3}{4}$

ANSWERS - COMPLETE MATHS – PART 39

3. Correct order of size is: $\frac{1}{2}$ $\frac{7}{12}$ $\frac{4}{6}$ $\frac{3}{4}$

4. Correct order of size is: $\frac{1}{4}$ $\frac{3}{10}$ $\frac{3}{8}$ $\frac{2}{5}$

5. Correct order of size is: $\frac{1}{6}$ $\frac{2}{9}$ $\frac{1}{3}$ $\frac{2}{3}$

6. Correct order of size is: $\frac{1}{8}$ $\frac{1}{4}$ $\frac{1}{3}$ $\frac{9}{24}$

7. Correct order of size is: $\frac{4}{1}$ $\frac{1}{3}$ $\frac{2}{5}$ $\frac{7}{15}$

8. Correct order of size is: $\frac{2}{3}$ $\frac{3}{4}$ $\frac{5}{6}$ $\frac{9}{10}$

Pie charts and percentages

1/ English - half or 1/2
Sport – eighth or 1/8
Maths – eighth or 1/8
Science – quarter or 1/4

2/ English – 50%
Sport – 12.5%
Maths – 12.5%
Science – 25%

3/ a 20
b 5
c 5
d 10

Equations revision

(Answers from left to right, top to bottom)

1. 9
2. 4
3. 1
4. 15
5. 5
6. 8
7. 8
8. 5
9. 5
10. 14
11. 10
12. 6

13. 5
14. 6
15. 15
16. 1
17. 3
18. 2
19. 6
20. 5
21. 2
22. 7

Equations Game

1.	$x \div 3 = 4$	$x=12$
2.	$x+3=5$	$x=2$
3.	$x+6=11$	$x=5$
4.	$x-3=1$	$x=4$
5.	$14-x=7$	$x=7$
6.	$x+7=13$	$x=6$
7.	$3+x=12$	$x=9$
8.	$x+8=9$	$x=1$
9.	$4x=20$	$x=5$
10.	$15-x=7$	$x=8$
11.	$x-4=1$	$x=5$
12.	$x \div 2=5$	$x=10$
13.	$\frac{x}{5}=3$	$x=15$
14.	$x+8=16$	$x=8$
15.	$10-x=5$	$x=5$
16.	$x \div 8=2$	$x=16$
17.	$4x=16$	$x=4$
18.	$11-x=5$	$x=6$
19.	$5x=25$	$x=5$
20.	$\frac{x}{3}=5$	$x=15$
21.	$3x=9$	$x=3$
22.	$x-1=5$	$x=6$
23.	$8x=16$	$x=2$
24.	$x+5=9$	$x=4$

Maths Type 2 Paper 5

Page 1

1. 2082
2. 5/6
3. 96
4. 8.65
5. a. 1.356 b. 0.27
   c. 9 2/3 d. 10725
6. 3.8km
7. 1.5kg or 1500g

Page 2

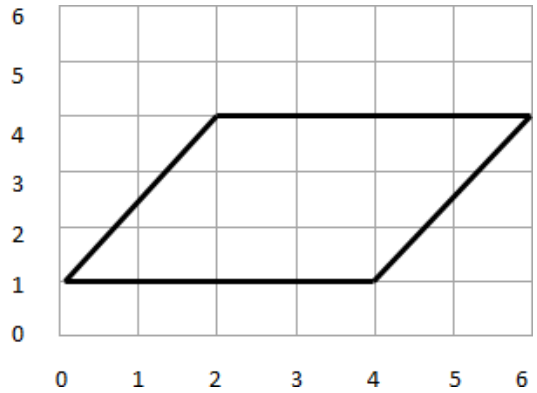
8. a. 3 b. 35
9. 37
10. 12m

Page 3

11. 55
12. 250ml
13. 3kg
14. D
15. a. 40% b. 8 c. 1:1
16. B and C

Page 4

17. 5.03pm
18. 1600
19. 39, 36
20. a.



b. 0

ANSWERS - COMPLETE MATHS – PART 39

Page 5

21. 12°
22. -35°C
23. 40.4°C
24. 51°
25. Square 400cm²

Page 6.

26. 74
27. a 1 day b 3 days
    c 4 months d 33 months
28. a 3 minutes b 20 km/h c 12 mph
29. 7 and 6

Page 7

30. 10°
31. 19°C
32. 1kg
33. 29
34. 6 litres

Core Skills 8

1. 36
2. 49.8
3. 84
4. 27Y + 27Z + 54X
   (or $\frac{1}{2}$Y + $\frac{1}{2}$Z + X)
5. 10 vases
6. 35.7
7. 1900
8. 34090
9. 5.88 m
10. £7.75

Maths Reasoning

a) 00:33:40 1 January 2020
b) 09:40:00 2 January 2020
c) 04:00 25 March 2020
d) 18:00 13 July 2025
e) 12:00 18 September 2058

Tip - 2020 is a leap year.

A leap year occurs once every 4 years when an extra day (29th February) is added on to the calendar to make up for the fact that the Earth's orbit around the sun actually takes 365 and a quarter days, not 365. A handy way to remember when the leap years fall is that they are always the same year as the summer Olympic Games (which also occur once every 4 years)!

Non-Verbal Reasoning

Like Shapes

1. **A** - (Three changes occur. The original innermost shape becomes the outermost shape. The original outer shape becomes the middle shape. The original black shape becomes the white innermost shape. Therefore the correct answer must be a diamond outer shape, pentagon middle shape and flag inner shape.)
2. **D** - (The first shape has one more side than the second shape. The black circle also moves to the opposite side of the shape and changes colour. Therefore the correct answer must be a two sided shape with a black circle at the bottom.)
3. **C** – (The first and third shapes in the sequence swap positions.)

Odd One Out

1. **A** – (All options have more triangles than circles apart from A.)
2. **B** – (All options share the same pattern of thick, thin, thick concentric rings, apart from B which has thin, thick, thick concentric rings.)
3. **B** – (All options have a quadrilateral as the inner shape apart from B.)

Rotations

1. **A** – (A is a rotation of 90 degrees anti-clockwise.)
2. **C** – (C is a rotation of approximately 135 degrees clockwise.)
3. **B** – (B is a rotation of approximately 135 degrees anti-clockwise.)

Code Breaker

1. **B** - (You can work out that G represents the outer shape being a hexagon as there are two of them. Therefore J stands for the outer shape being a rectangle. You can work out that Q stands for the inner triangle being shaded grey as there are two of them. Therefore O stands for the inner triangle being white. Therefore the answer is JO.)